Predicting In-Vehicle Coupon Acceptance with Machine Learning

ISBA2403 - Winter 2026 - Group 9

1. Background

1.1 Project Overview

In many real-world scenarios, coupons are still distributed in a broad and untargeted way, leading to low acceptance rates and unnecessary marketing waste. This issue becomes even more pronounced in driving contexts, where users have limited attention and make quick decisions, making irrelevant offers more likely to be ignored. Our project focuses on improving coupon targeting by predicting whether a user will accept a coupon in an in-vehicle setting. By leveraging user behavior, trip context, and coupon characteristics, our goal is to enable more precise and efficient coupon delivery.

1.2 Machine Learning Task

We formulate this problem as a binary classification task, where the objective is to predict whether a coupon will be accepted or rejected. The model takes in multiple types of features, including behavioral patterns, contextual information, coupon attributes, and demographic variables. The output is a predicted probability of acceptance, which can be used to rank users and support decision-making in coupon allocation.

2. Data Overview

2.1 Dataset Description

- **Source**: UCI In-Vehicle Coupon Recommendation dataset
- **Observations**: 12,684 driving scenarios

Each observation represents a moment in which a user receives a coupon while driving, along with information about their current context, past behavior, and personal characteristics.

2.2 Feature Description

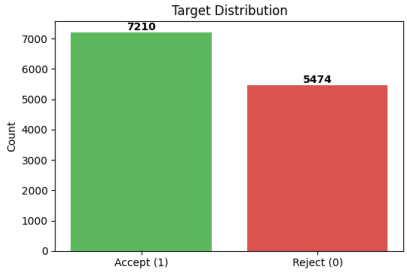
The dataset includes a mix of categorical and numerical variables, covering user behavior, trip context, demographics, and coupon attributes. The target variable is binary, indicating whether the coupon is accepted or rejected.

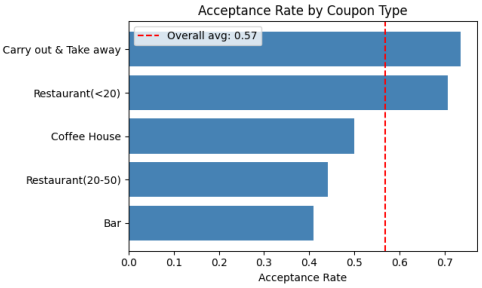
There are 25 features in total and 1 target variable (Y). The features can be grouped into four categories:

- Context Features: Time, weather, passenger
- Behavioral Features: Bar, CoffeeHouse, CarryOut, Restaurant (<20), Restaurant (20-50),
- Demographic Features: Age, income, children, marital status
- Coupon Features: Type, expiration

2.3 Exploratory Data Analysis (EDA)

2.3.1 Dataset Overview





The target variable **Y** indicates whether a driver accepts a coupon (1) or not (0).

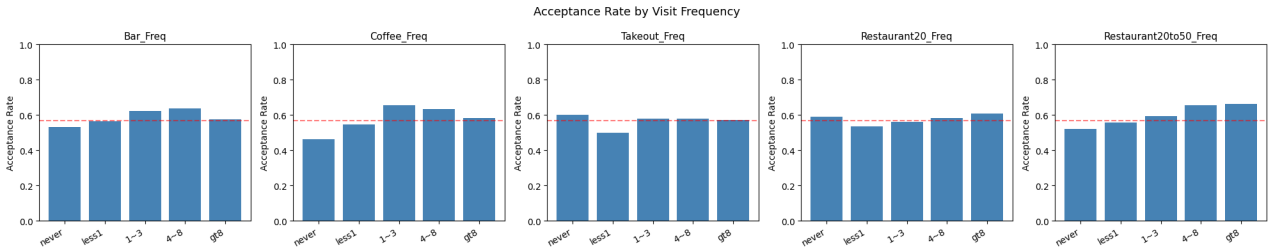
- Accept (1): **7,210 (56.8%)**
- Reject (0): **5,474 (43.2%)**

The dataset is moderately balanced, making standard classification metrics (e.g., ROC-AUC, F1-score) appropriate without requiring imbalance correction.

2.3.2 Key Drivers of Coupon Acceptance

EDA reveals that coupon acceptance is driven by the interaction of four main factors:

(1) Behavioral Patterns



Acceptance increases from low to moderate visit frequency (1–8 times/month), then stabilizes or declines.

- Moderate users → highest responsiveness
- Low frequency → low relevance
- High frequency → diminishing returns

Insight: Behavioral familiarity combined with flexibility drives engagement.

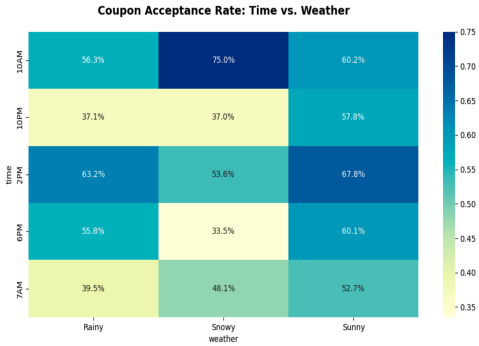
(2) Coupon Type

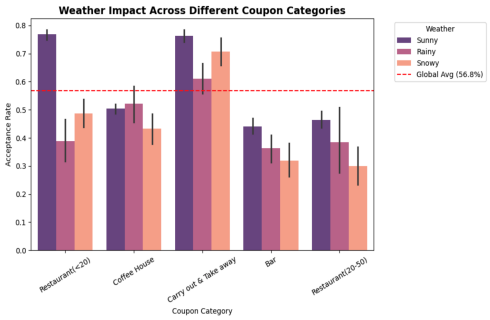
Acceptance varies significantly across coupon categories:

- High: CarryOut, Restaurant (<$20)
- Moderate: Coffee House, Restaurant ($20–50)
- Low: Bar

Insight: Low-friction, convenient coupons are more aligned with in-transit decision-making.

(3) Contextual Factors (Time & Weather)





- Highest acceptance: mid-day (10AM–2PM), especially in good weather
- Lower acceptance: early morning, late night, and adverse weather

Insight: Favorable conditions increase flexibility, while constraints reduce likelihood of redemption.

(4) Expiration Time

- 1-day expiration: 62.6%
- 2-hour expiration: 49.6%

Insight: Longer expiration reduces decision pressure and increases acceptance.

2.3.3 Demographic Effects (Secondary Role)

- Younger and single users show higher acceptance
- Income and education show weak or inconsistent effects

Insight: Demographics provide limited predictive power compared to behavioral features.

2.3.4 Key Insight Summary

Coupon acceptance is highest when:

- Low-friction coupons are delivered to behaviorally aligned users
- Under favorable conditions with sufficient time flexibility

Overall, behavioral patterns dominate, while coupon type, context, and timing act as interacting modifiers.

2.3.5 Implication for Modeling

The EDA findings suggest that:

- Coupon type captures baseline attractiveness
- Context and expiration reflect situational constraints
- Behavioral variables have stronger predictive power compared to demographic features

These elements allow models to learn who the user is (behavior), what is offered (coupon types), and under what conditions (context and time). These factors guide feature selection and model design in the next section.

2.4 Data Preprocessing

To prepare the dataset for modeling, we applied the following key preprocessing steps:

Missing Data Handling: Features with excessive missing values (e.g., car) were removed. Missing behavioral values were treated as "no prior visits."
Feature Transformation: Behavioral and ordinal variables (e.g., visit frequency, age, income) were converted into numerical representations to enable modeling.
Feature Cleaning: Redundant and near-constant features were removed, and collinear variables were consolidated to improve model stability.
Categorical Encoding: Categorical variables were transformed using one-hot encoding.
Feature Engineering: To improve model performance and ensure feature validity, the following feature engineering steps were applied

- **Removal of Non-informative Features:** The variable *toCoupon_GEQ5min* was dropped due to near-constant values (>99% identical), providing no predictive value.
- **Handling of Collinearity:** A new feature, *direction_consistency*, was created as the difference between *direction_same* and *direction_opp.* The original variables were removed due to perfect collinearity (*direction_same + direction_opp = 1*).

All feature engineering and preprocessing steps were implemented within a train-only pipeline to ensure leakage-safe modeling.

3. Model

3.1 Modeling Strategy Overview

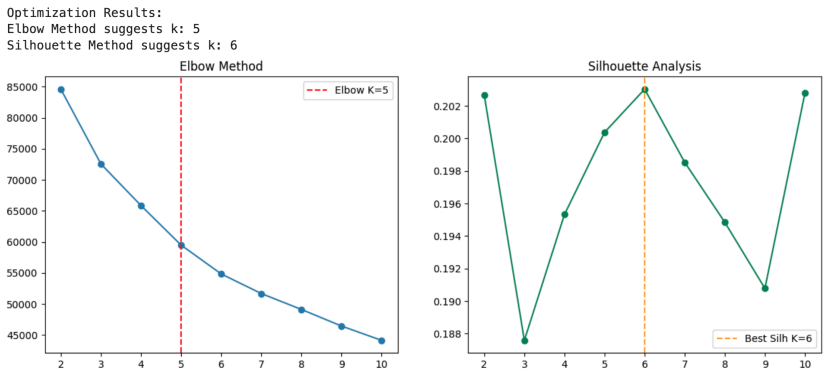
This project combines both supervised and unsupervised learning approaches. Supervised models are used to predict whether a coupon will be accepted, while clustering is used to capture user behavioral segments and enhance model performance.

3.2 Unsupervised Learning - Clustering

3.2.1 Objective

A single global model cannot fully capture these variations. Clustering is therefore necessary to identify latent user segments, enabling the model to capture differences in lifestyle, responsiveness, and sensitivity to coupon types. These segments are incorporated into the model to capture differences across users that are not directly observable from individual features

3.2.2 Methodology



We apply K-Means clustering using selected demographic and behavioral features, including age, income, presence of children, and frequency of visiting bars, coffee houses, and restaurants. Contextual and coupon-related variables are excluded to ensure that clusters represent stable user characteristics rather than situational factors. The number of clusters (K=5) is chosen based on interpretability and stability, balancing meaningful segmentation with model simplicity. Clustering is performed on the training set only, and the resulting centroids are used to transform the test set to avoid data leakage. Cluster information is incorporated into the model through one-hot encoded *cluster labels* and *distance-to-centroid* features.

3.2.3 Cluster Profiles

Five distinct user segments emerge:

- **Young Singles (≈22%)** - moderate activity, average acceptance
- **Social Food Lovers (≈5%)** - high dining frequency, high acceptance
- **Family Takeout (≈14%)** - convenience-driven, stable acceptance
- **Conservative Families (≈23%)** - low activity, lowest acceptance
- **Young Bar Hoppers (≈6%)** - high bar activity, highest acceptance

These segments differ in both behavior and responsiveness, supporting the use of segmentation in modeling and targeted coupon delivery.

3.2.4 Leakage-Safe Cluster Feature Integration

To prevent data leakage, a 70/30 stratified train–test split was applied before any fitting. All subsequent fitting, including the preprocessing pipeline, cluster scaler, and KMeans model, was performed exclusively on the training set. The resulting centroids were then used to transform the test set, ensuring no test information influenced the learned representations. Cluster-based features include:

- Cluster labels (one-hot encoded)
- Distance to cluster centroids

Enhanced feature set = **preprocessed features + cluster features**

3.3 Classification Model 1 - Logistic Regression

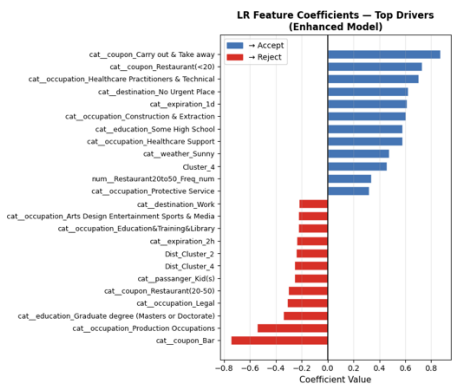
A baseline model and an enhanced model with clustering was created for the purpose of understanding the acceptance rate.

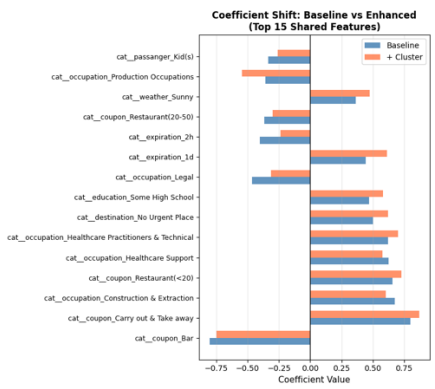
Both models performed similarly, converging at **AUC ~0.72** and **F1 ~0.73**, which represents a moderate but not strong predictive ability. Adding cluster features (Enhanced model) produced negligible gains, only +0.0015 AUC and +0.0012 F1, meaning the segmentation step added complexity without meaningful predictive value.

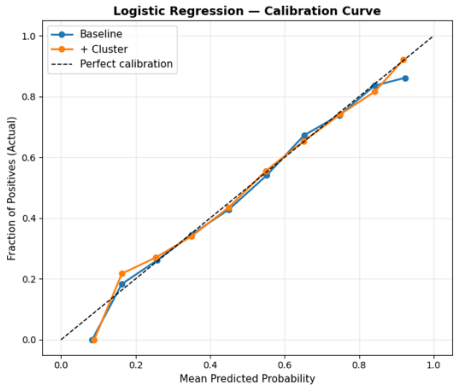
Feature coefficients chart show that coupon type is the dominant driver — Carry out & Take away and Restaurant (<20) are the strongest accept signals, while Bar coupons are the strongest reject signal, nearly equal in magnitude. Context also matters: heading to No Urgent Place and a 1-day expiration both push toward acceptance, while 2-hour expiration and going to Work push toward rejection, this shows that giving people urgency pushes acceptance rate lower in the world of coupons. Occupation plays a surprisingly large role, with healthcare workers consistently accepting and production/legal occupations rejecting.

Coefficient shift graph confirms that adding cluster features barely re-distributed the model's learned weights — nearly every bar pair is almost identical in length. The cluster features did not absorb or replace existing feature signals. The only features showing any visible shift are cat_occupation_Production Occupations and expiration/weather features, but the changes are minor.

Calibration curve shows that both models track the perfect calibration diagonal very closely across the full probability range, with only minor deviation at the low end (~0.08–0.15) . This means the model's probability outputs are trustworthy with a predicted 70% acceptance probability.







Overall, the model is well-calibrated but limited in discriminative power**.** AUC of 0.72 means the model correctly ranks a random accept above a random reject about 72% of the time, but leaving meaningful room for improvement.

Coupon type and context are the real drivers. If this model were used in a car company to implement such coupon service in the central screen, the actionable insight is clear: carry-out and cheap restaurant coupons in low-urgency contexts with longer expiry windows are most likely to be accepted. Bar coupons and short-window offers under work commute conditions are most likely to be rejected.

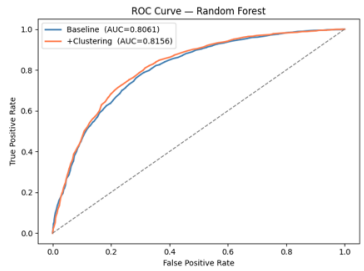
Clustering did not add predictive value for logistic regression model. Despite Cluster_4 showing a positive coefficient, the overall cluster feature set failed to improve the model meaningfully.

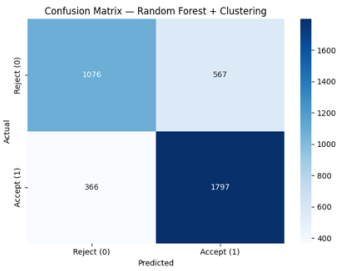
Logistic regression has reached its ceiling for predicting acceptance. The remaining ~28% AUC gap suggests non-linear interactions between features — for example, the effect of occupation likely interacts with coupon type, which a linear model cannot capture.

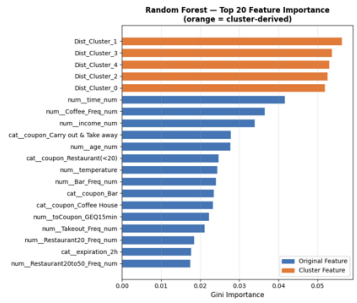
3.4 Classification Model 2 - Random Forest

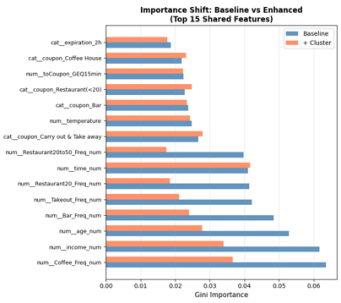
Random Forest is applied to capture nonlinear relationships and feature interactions that cannot be modeled by Logistic Regression. By aggregating multiple decision trees, it improves predictive performance while reducing overfitting compared to a single tree.

The model is trained under both baseline (original features) and enhanced (with clustering features) settings. Compared to Logistic Regression, Random Forest achieves a substantial performance improvement, with AUC increasing from **0.8061** to **0.8156** after incorporating clustering features **(+0.0095)**.









Feature importance analysis shows a shift from individual features to interaction-driven signals. In particular, cluster distance features play a more prominent role, indicating that user segmentation helps define decision boundaries in tree-based models. At the same time, the importance of demographic variables decreases, suggesting that behavioral patterns are more informative than static user attributes.

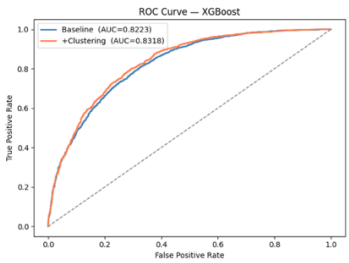
Unlike Logistic Regression, Random Forest can leverage segmentation information more effectively, as it can split on cluster-related features in specific regions of the feature space. This highlights that user identity (who the user is) becomes more relevant when nonlinear interactions are considered.

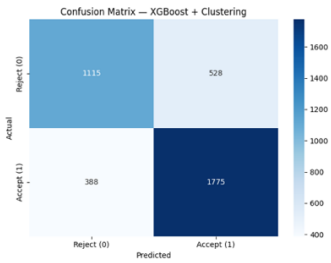
Overall, Random Forest demonstrates strong improvements over the linear baseline and confirms the presence of nonlinear relationships in coupon acceptance behavior. However, its performance remains below XGBoost, suggesting that further gains require more flexible modeling of feature interactions.

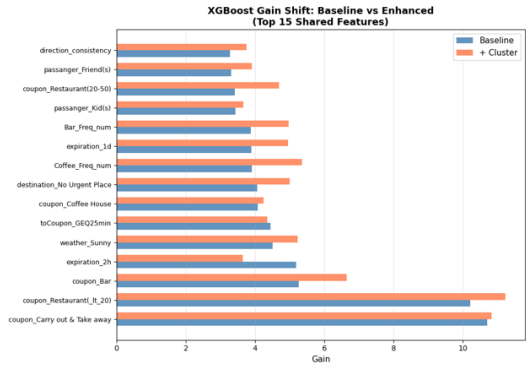
3.5 Classification Model 3 - XGBoost

XGBoost is used as the most advanced model to capture nonlinear relationships and feature interactions in coupon acceptance behavior. Compared to Logistic Regression and Random Forest, it leverages gradient boosting to iteratively reduce prediction errors, making it well-suited for structured tabular data.

The model is trained under two settings: a baseline version using original features, and an enhanced version incorporating clustering features (persona indicators and distances to cluster centroids). Hyperparameters are tuned using RandomizedSearchCV with 5-fold stratified cross-validation, optimizing for ROC-AUC to align with the ranking-based business objective.

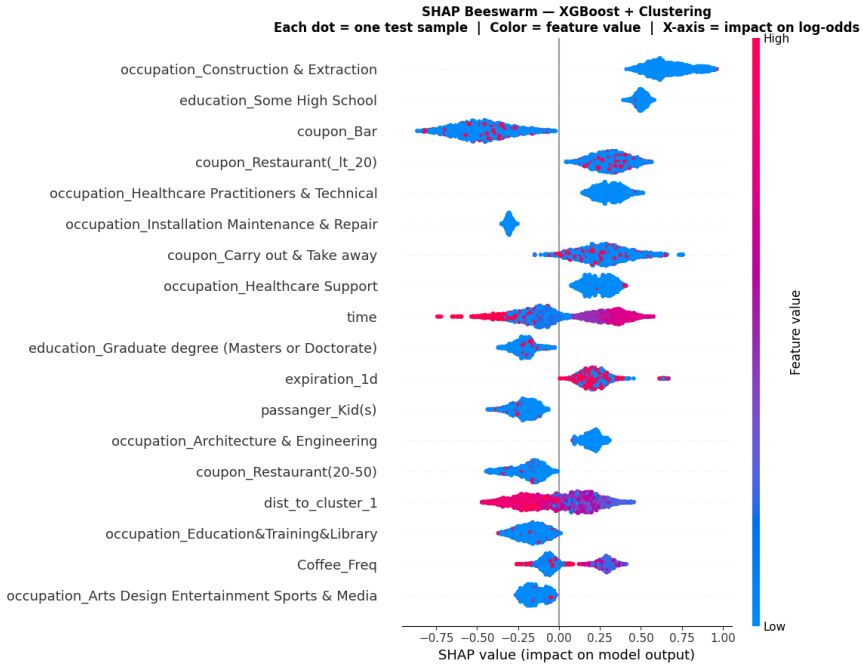






XGBoost achieves the best performance among all models. The baseline model reaches an AUC of **0.8223**, which improves to **0.8318** after adding clustering features (+0.0095). Cross-validation AUC (0.8335) closely matches test performance, indicating strong generalization without overfitting. The consistent improvement across models suggests that user segmentation provides meaningful predictive signals.

Feature importance analysis shows that coupon type remains the strongest global driver. However, unlike Logistic Regression, XGBoost captures interactions between coupon type, user behavior, and contextual factors. Cluster features are not uniformly important but are selectively used, indicating that segmentation matters only under certain conditions.



SHAP analysis further reveals that carry-out coupons consistently drive acceptance, while bar coupons tend to be rejected unless aligned with the appropriate user segment. Contextual effects are also evident—for example, late-night scenarios reduce acceptance rates. Additionally, distance to certain clusters (e.g., Social Food Lovers) influences predictions, confirming that behavioral segmentation contributes to model decisions.

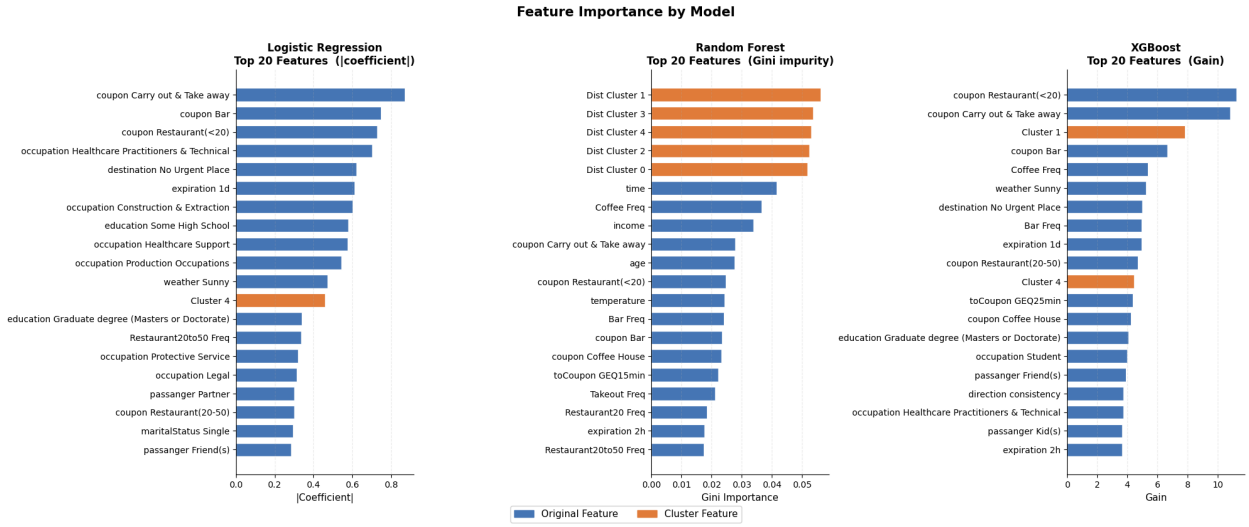
Overall, XGBoost outperforms other models by capturing the interaction structure underlying user decisions. Coupon acceptance is not driven by individual features alone, but by the alignment of user type, context, and offer.

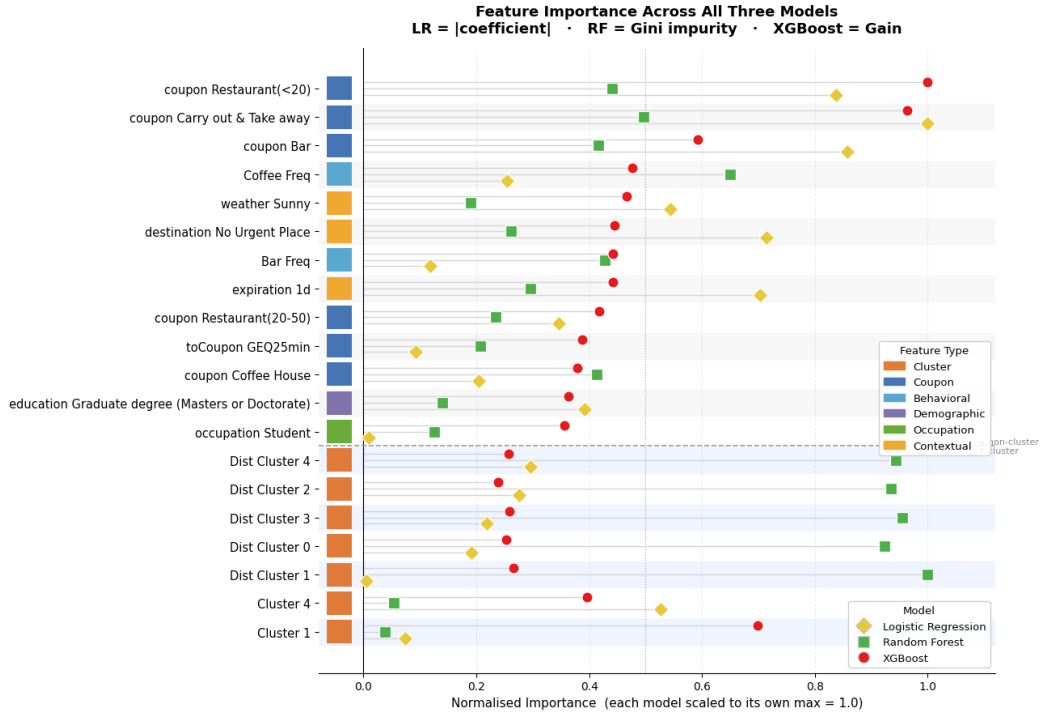
3.6. Model Summary

3.6.1 Test Performance Comparison

Model	Test AUC	Test F1
Logistic Regression	0.722782	0.730228
Logistic Regression + Clustering	0.724297	0.731441
Random Forest	0.806099	0.786378
Random Forest + Clustering	0.815624	0.793903
XGBoost	0.822298	0.787010
XGBoost + Clustering	0.831769	0.794895

3.6.2 Feature Importance





The two charts above present feature importance results across all three models in two complementary views. The first chart (top) compares feature importance side by side per model using their native metrics: absolute coefficient for Logistic Regression, Gini impurity for Random Forest, and Gain for XGBoost. Orange bars indicate cluster-derived features; blue bars indicate original features. The second chart (bottom) presents a unified lollipop view where each model's scores are normalized to its own maximum, enabling relative cross-model comparison on a common scale.

Logistic Regression: Coupon type dominates, with *coupon Carry out & Take away* and *coupon Restaurant(<20)* holding the largest positive coefficients, while *coupon Bar* is the strongest reject signal. Occupation features also play a notable role, ranking unexpectedly high among top coefficients.

Random Forest: All five cluster distance features occupy the top five positions by Gini impurity, far outranking original features. This reflects the tendency of tree-based models to favor continuous distance scores as efficient split criteria.

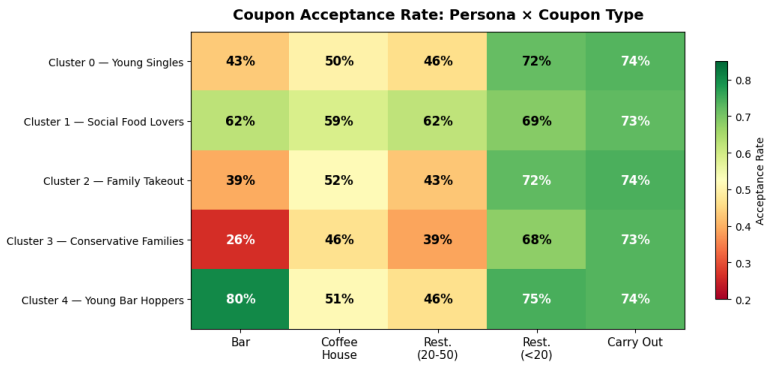
XGBoost: Coupon type leads on Gain and Cluster_1 (Social Food Lovers) ranks third, the only cluster feature within the top five by Gain. XGBoost integrates both coupon type and contextual features (weather, destination, expiration) alongside cluster signals, making it the only model that captures all three dimensions simultaneously. This breadth of signal integration is consistent with XGBoost achieving the highest overall AUC.

4.Prescriptive Analysis

4.1 Segmentation Analysis

Users are grouped into five behavioral segments: Young Singles, Social Food Lovers, Family Takeout, Conservative Families, and Young Bar Hoppers.

Acceptance varies significantly across segments. Social Food Lovers and Young Bar Hoppers are the most responsive, while Conservative Families show the lowest engagement. This suggests that a uniform coupon strategy is inefficient, and targeting by segment is necessary.

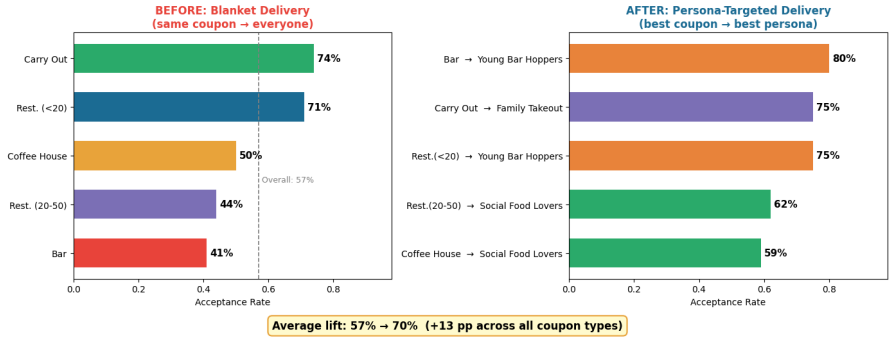


4.2 Targeting — Right Coupon, Right Persona

The most critical insight from this project is: Who you send to matters more than what you send. The same coupon can produce dramatically different outcomes:

- Bar coupon → Conservative Families: **26% acceptance**
- Bar coupon → Young Bar Hoppers: **80% acceptance**

By moving from blanket strategy to persona-targeted delivery, average acceptance improves from 57% → 70% (+13%).

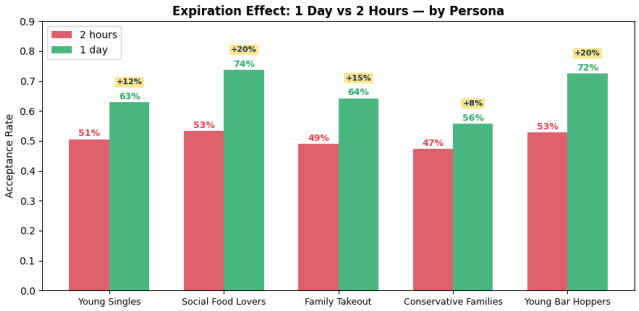


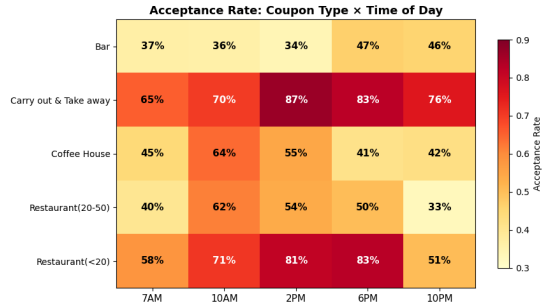
4.3 Timing — Longer Decision, Precise Delivery

Coupon effectiveness varies significantly depending on timing and contextual factors.

- Longer expiration (1-day) > short-term urgency (2-hour)

- Effectiveness varies by time of day (e.g., coffee morning, bar evening)
- Coffee performs better in rainy conditions

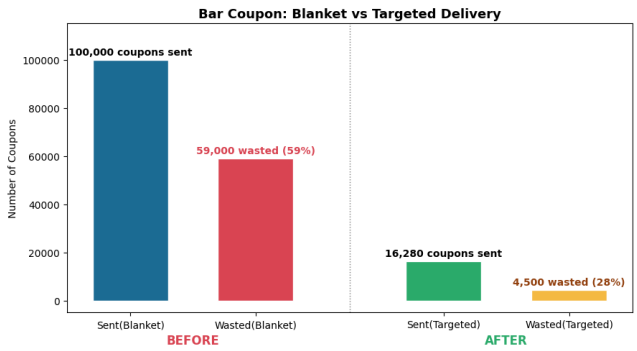




4.4 Triage — Send Fewer, Earn More

We simulate a triage strategy on bar coupon

- Blanket: 100,000 coupons sent, 59,000 wasted
- Targeted: 16,280 coupons sent, 4,500 wasted
- 31% waste reduction, 41%–72% ROI increase



At scale: 100K coupons/day × $0.5 × 31% reduction = ~$16K/day savings (Bar segment alone)

5. Conclusion

5.1 Summary of Findings

Our project shows that coupon acceptance is driven by the interaction of behavioral patterns, contextual factors, and user segmentation. While coupon type provides a strong baseline signal, differences across users and situations lead to large variation in outcomes. By combining predictive modeling with segmentation, we demonstrate that targeted delivery—matching the right coupon to the right user at the right time—can significantly improve acceptance rates while reducing unnecessary marketing cost.

5.2 Limitations

Our analysis has several limitations. The dataset does not include real-time signals such as location or user intent, which are likely to influence decision-making in practice. In addition, clustering is static and may not fully capture dynamic changes in user behavior. External factors such as brand preference or pricing are also not considered.

5.3 Future Work

Future work can focus on incorporating real-time contextual data and modeling user behavior over time using sequential approaches. In addition, online A/B testing can be used to validate the effectiveness of the proposed strategies. Ultimately, this can support the development of a scalable, end-to-end recommendation system for coupon delivery.